热点读者画像：技术细节与源码连接位置

按数据流阅读：离线生成聚合 → 热点排序 → 批量画像 → 模型输入 → 保存与页面

01 离线确认读者

build_package 按 tenant/news/window 对 detail_view 用户去重。

/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/build.py:359
/Users/shi/Project/myProject/NewsAgent/agent/app/behavior_tags/build.py:88

02 计算访问前兴趣与统计

_RollingState 维护30天/24小时兴趣；_dimension 算人数、占比与 lift。

/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/build.py:219
/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/build.py:251

03 入口与运行身份

页面运行入口冻结 reader context；ActiveBundle 校验窗口与批准的 Prompt。

/Users/shi/Project/myProject/NewsAgent/agent/examples/local_simulation.py:223
/Users/shi/Project/myProject/NewsAgent/agent/app/services/active_bundle_hot_news.py:194

04 热点排名接画像补全

先 rank，再 enrich；_fetch_reader_profiles 每20篇构造一批 Query。

/Users/shi/Project/myProject/NewsAgent/agent/app/services/hot_news_orchestration.py:283
/Users/shi/Project/myProject/NewsAgent/agent/app/analytics/hot_news_enrichment.py:198

05 在线聚合检索

Service 做缓存/请求合并；Local 用参数化索引查，企业经固定 HTTPS。

/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/service.py:138
/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/local.py:118
/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/client.py:114

06 将聚合送进模型

InputBuilder 复制 reader_profile_summary；Runner 调结构化模型 Port。

/Users/shi/Project/myProject/NewsAgent/agent/app/services/hot_news_analysis.py:109
/Users/shi/Project/myProject/NewsAgent/agent/app/model_runtime/hot_news.py:119

07 保存并返回报告

Activity 保存完成结果；GET查询从快照投影 summary 与 trace。

/Users/shi/Project/myProject/NewsAgent/agent/app/activities/hot_news.py:175
/Users/shi/Project/myProject/NewsAgent/agent/app/services/hot_news_run_store.py:67
/Users/shi/Project/myProject/NewsAgent/agent/app/services/hot_news_query.py:192

08 页面展示与策略复用

榜单把 summary/trace 传入画像组件；推送策略输入复用同一聚合。

/Users/shi/Project/myProject/NewsAgent/agent/frontend/src/components/hotnews/HotNewsRankTable.vue:112
/Users/shi/Project/myProject/NewsAgent/agent/frontend/src/components/hotnews/NewsReaderProfilePanel.vue:26
/Users/shi/Project/myProject/NewsAgent/agent/app/services/push_strategy.py:58

装配、独立接口与边界

装配：/Users/shi/Project/myProject/NewsAgent/agent/app/reader_profiles/factory.py:52 | 配置：/Users/shi/Project/myProject/NewsAgent/agent/app/config.py:62
独立 POST 查询：/Users/shi/Project/myProject/NewsAgent/agent/app/api/reader_profiles.py:30；热点内部直接调用同一 Service。
测试：test_reader_profile_*.py；企业百万/千万级 SLA、动态 Schedule 冻结和整栈页面仍需验收。